# Correlation and Testing

Questions 1–19

**2.46 — Question 1. Tables of critical values are on page 356** Describe what is meant by the significance level.

**2.46 — Question 2.** In each of the following cases, $r$ is the sample correlation coefficient. Determine whether you reject $H_0$ or fail to reject $H_0$

a) $H_0 : \rho = 0$, $H_1 : \rho > 0$
The test statistic is $r = 0.638$ and the $p$-value is 0.0444, where $n = 8$ and the significance level is 5%

b) $H_0 : \rho = 0$, $H_1 : \rho < 0$
The test statistic is $r = -0.456$ and the $p$-value is 0.0681, where $n = 12$ and the significance level is 5%

**2.46 — Question 3.** In each of the following cases, $r$ is the sample correlation coefficient. Determine whether you reject $H_0$ or fail to reject $H_0$

a) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.487$
Critical value for $n = 14$ for a one-tailed test at a 5% significance level is 0.4575

b) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.628$
Critical value for $n = 7$ for a one-tailed test at a 5% significance level is 0.6694

c) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.593$
Critical value for $n = 20$ for a one-tailed test at a 1% significance level is 0.5155

d) $H_0 : \rho = 0$, $H_1 : \rho > 0$
$r = 0.213$
Critical value for $n = 60$ for a one-tailed test at a 10% significance level is 0.1678

**2.46 — Question 4.** In each of the following cases, $r$ is the sample correlation coefficient. Determine whether you reject $H_0$ or fail to reject $H_0$

a) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.453$
Critical value for $n = 15$ for a one-tailed test at a 5% significance level is 0.4409

b) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.829$
Critical value for $n = 7$ for a one-tailed test at a 5% significance level is 0.6694

c) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.306$
Critical value for $n = 25$ for a one-tailed test at a 1% significance level is 0.4622

d) $H_0 : \rho = 0$, $H_1 : \rho < 0$
$r = -0.197$
Critical value for $n = 40$ for a one-tailed test at a 10% significance level is 0.2070

**2.46 — Question 5.** State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho > 0$
Critical value for $n = 19$ for a one-tailed test at a 0.5% significance level is 0.5751

**2.46 — Question 6.** State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho < 0$
Critical value for $n = 16$ for a one-tailed test at a 1% significance level is 0.5742

**2.46 — Question 7.** In each of the following cases, $r$ is the sample correlation coefficient. Determine whether you reject $H_0$ or fail to reject $H_0$

a) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = 0.658$
Critical value for $n = 9$ for a two-tailed test at a 5% significance level is 0.6664

b) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = -0.567$
Critical value for $n = 11$ for a two-tailed test at a 5% significance level is 0.6021

c) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = 0.621$
Critical value for $n = 16$ for a two-tailed test at a 1% significance level is 0.6226

d) $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
$r = -0.248$
Critical value for $n = 50$ for a two-tailed test at a 10% significance level is 0.2353

**2.46 — Question 8.** State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho \neq 0$
Critical value for $n = 7$ for a two-tailed test at a 2.5% significance level is 0.7545

**2.46 — Question 9.** State the critical region for the following test:
$H_0 : \rho = 0$, $H_1 : \rho \neq 0$
Critical value for $n = 80$ for a two-tailed test at a 5% significance level is 0.2199

**2.46 — Question 10.** Hannah believes there is a positive correlation between the amount of sugar consumed daily by adults and their blood glucose levels. She collects a random sample of 28 adults and finds the sample's correlation coefficient to be 0.382. Carry out a hypothesis test at the 5% significance level to investigate whether Hannah's belief is supported.

**2.46 — Question 11.** Kyle is a social scientist who wants to explore if there is a negative correlation between the number of hours spent on social media each day and self-reported feelings of loneliness in teenagers. He collects a random sample of 70 teenagers and finds the sample's correlation coefficient to be $-0.297$. Carry out a hypothesis test at the 0.5% significance level to investigate whether there is evidence to support a negative correlation.

**2.46 — Question 12.** Hamza is a biologist who is investigating whether there is a negative correlation between the water temperature they grow up in and their growth rate. Hamza wants to conduct a hypothesis test at the 1% significance level. He collects a random sample of 40 of a certain type of fish and finds the sample's correlation coefficient to be $-0.364$. Hamza's hypothesis test is shown below.

*Let $\rho$ be the sample correlation coefficient between the water temperature and the growth rate of the fish.*

$H_0 : \rho = 0$

$H_1 : \rho < 0$

$r = -0.364$

*The critical value for $n = 40$ for a one-tailed test at a 1% significance level is* 0.3665

$-0.364 < 0.3665$ *so we reject* $H_0$

*There is sufficient evidence to suggest that there is a negative correlation between the water temperature and the growth rate of the fish.*

a) Identify the first two errors made by Hamza in his hypothesis test.

b) Correct his test.

**2.46 — Question 13.** Aida is an environmental scientist who wants to determine if there is some correlation between the levels of air pollution and numbers of adults with respiratory illnesses. She collects data from 50 randomly selected neighbourhoods and finds the sample correlation coefficient to be 0.209. Carry out a hypothesis test at the 5% significance level to investigate whether there is any evidence to support there being some correlation.

**2.46 — Question 14.** Kaylee is a nutritionist who wants to determine if there is any correlation between the daily intake of fruits and vegetables in adults and their body mass index (BMI). She collects data from 100 randomly selected adults and finds the sample correlation coefficient to be $-0.301$. Carry out a hypothesis test at the 1% significance level to investigate whether there is any evidence to suggest a correlation.

**2.46 — Question 15.** Jo is a psychologist who is investigating whether there is any correlation between the number of hours of sleep college students get each night and their stress levels, as measured by a standardised test. Jo wants to conduct a hypothesis test at the 5% significance level. She collects data from 28 randomly selected college students and finds the sample correlation coefficient to be 0.482. Jo's hypothesis test is shown below.

*Let $\rho$ be the population correlation coefficient between the number of hours of sleep college students get each night and their stress levels.*

$H_0 : \rho = 0$

$H_1 : \rho \neq 0$

$r = 0.482$

*The critical value for $n = 28$ for a two-tailed test at a 5% significance level is* 0.3172

$0.3172 < 0.482$ *so we fail to reject* $H_0$

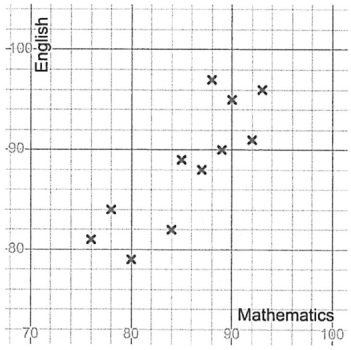
*There is insufficient evidence to suggest that there is some correlation between the number of hours of sleep college students get each night and their stress levels.*

a) Identify the first two errors made by Jo in her hypothesis test.

b) Correct her test.

**2.47 — Question 1. Tables of critical values are on page 356** Richard is investigating whether there is an association between the scores students achieve in a Mathematics test and in an English test. He takes a random sample of 11 students and records their scores in each subject. The table below shows the data, and below that is a scatter graph to illustrate the data.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Mathematics | 88 | 85 | 90 | 78 | 92 | 87 | 84 | 80 | 93 | 76 | 89 |
| English | 97 | 89 | 95 | 84 | 91 | 88 | 82 | 79 | 96 | 81 | 90 |




Richard decides to carry out a Spearman's Rank hypothesis test to investigate whether there is a positive association between the scores in the two subjects, using a 5% significance level.

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

a) Use the formula above to show that $r_s = 0.845$ to 3sf

**2.47 — Question 1 (continued).** Richard writes out the first part of his hypothesis test correctly, as shown below:
$H_0$: *There is no association between Mathematics test scores and English test scores in the population*
$H_1$: *There is a positive association between Mathematics test scores and English test scores in the population*
$r_s = 0.845$
*The critical value for* $n = 11$ *for a one-tailed test at a 5% significance level is* 0.5364
$0.845 > 0.5364$

b) Select Richard's correct conclusion:

| | |
|---|---|
| *Reject* $H_0$. *There is sufficient evidence to suggest that there is a positive association between Mathematics and English test scores in the population.* | *Fail to reject* $H_0$. *There is insufficient evidence to suggest that there is a positive association between Mathematics and English test scores in the population.* |

**2.47 — Question 2.** Beth is investigating whether there is a positive association between the length of a bird's wingspan and its tendency to migrate long distances. She measures the wingspans, in cm, and the maximum migration distances, in km, of a random sample of 12 different bird species. Beth correctly calculates the Spearman's Rank correlation coefficient to be 0.479. Conduct a test at the 5% level to investigate whether there is a positive association between the wingspan and migration distance.

**2.47 — Question 3.** Si is investigating whether there is a negative association between the air quality index (AQI) and the average life expectancy, in years, of people in different cities. He collects a random sample of 15 cities and correctly calculates the Spearman's Rank correlation coefficient to be $-0.654$. Conduct a test at the 1% level to investigate whether there is a negative association between the air quality index and average life expectancy.

**2.47 — Question 4.** Claire is investigating whether there is any association between the number of hours of sleep a person gets per night and their cognitive performance score on a standardised test. She collects a random sample of 22 individuals who were tested and correctly calculates the Spearman's Rank correlation coefficient to be 0.321. Conduct a test at the 10% level to investigate whether there is any association between the number of hours of sleep and cognitive performance score.

# 2.48 PMCC and Spearman's Critical Tables

Reference tables

### Critical Values for the Product Moment Correlation Coefficient

| 2-Tail | 20% | 10% | 5% | 2% | 1% | Sample |
|---|---|---|---|---|---|---|
| 1-Tail | 10% | 5% | 2.5% | 1% | 0.5% | size, $n$ |
| | 0.8000 | 0.9000 | 0.9500 | 0.9800 | 0.9900 | 4 |
| | 0.6870 | 0.8054 | 0.8783 | 0.9343 | 0.9587 | 5 |
| | 0.6084 | 0.7293 | 0.8114 | 0.8822 | 0.9172 | 6 |
| | 0.5509 | 0.6694 | 0.7545 | 0.8329 | 0.8745 | 7 |
| | 0.5067 | 0.6215 | 0.7067 | 0.7887 | 0.8343 | 8 |
| | 0.4716 | 0.5822 | 0.6664 | 0.7498 | 0.7977 | 9 |
| | 0.4428 | 0.5494 | 0.6319 | 0.7155 | 0.7646 | 10 |
| | 0.4187 | 0.5214 | 0.6021 | 0.6851 | 0.7348 | 11 |
| | 0.3981 | 0.4973 | 0.5760 | 0.6581 | 0.7079 | 12 |
| | 0.3809 | 0.4762 | 0.5529 | 0.6339 | 0.6835 | 13 |
| | 0.3646 | 0.4575 | 0.5324 | 0.6120 | 0.6614 | 14 |
| | 0.3507 | 0.4409 | 0.5140 | 0.5923 | 0.6411 | 15 |
| | 0.3383 | 0.4259 | 0.4973 | 0.5742 | 0.6226 | 16 |
| | 0.3271 | 0.4124 | 0.4821 | 0.5577 | 0.6055 | 17 |
| | 0.3170 | 0.4000 | 0.4683 | 0.5425 | 0.5897 | 18 |
| | 0.3077 | 0.3887 | 0.4555 | 0.5285 | 0.5751 | 19 |
| | 0.2992 | 0.3783 | 0.4438 | 0.5155 | 0.5614 | 20 |
| | 0.2914 | 0.3687 | 0.4329 | 0.5034 | 0.5487 | 21 |
| | 0.2841 | 0.3598 | 0.4227 | 0.4921 | 0.5368 | 22 |
| | 0.2774 | 0.3515 | 0.4133 | 0.4815 | 0.5256 | 23 |
| | 0.2711 | 0.3438 | 0.4044 | 0.4716 | 0.5151 | 24 |
| | 0.2653 | 0.3365 | 0.3961 | 0.4622 | 0.5052 | 25 |
| | 0.2598 | 0.3297 | 0.3882 | 0.4534 | 0.4958 | 26 |
| | 0.2546 | 0.3233 | 0.3809 | 0.4451 | 0.4869 | 27 |
| | 0.2497 | 0.3172 | 0.3739 | 0.4372 | 0.4785 | 28 |
| | 0.2451 | 0.3115 | 0.3673 | 0.4296 | 0.4705 | 29 |
| | 0.2407 | 0.3061 | 0.3610 | 0.4226 | 0.4630 | 30 |
| | 0.2070 | 0.2638 | 0.3120 | 0.3665 | 0.4026 | 40 |
| | 0.1843 | 0.2353 | 0.2787 | 0.3281 | 0.3610 | 50 |
| | 0.1678 | 0.2144 | 0.2542 | 0.2997 | 0.3301 | 60 |
| | 0.1550 | 0.1982 | 0.2352 | 0.2776 | 0.3060 | 70 |
| | 0.1448 | 0.1852 | 0.2199 | 0.2597 | 0.2864 | 80 |
| | 0.1364 | 0.1745 | 0.2072 | 0.2449 | 0.2702 | 90 |
| | 0.1292 | 0.1654 | 0.1966 | 0.2324 | 0.2565 | 100 |

### Critical Values for Spearman's Rank Correlation Coefficient

| 2-Tail | 10% | 5% | 2% | 1% | Sample |
|---|---|---|---|---|---|
| 1-Tail | 5% | 2.5% | 1% | 0.5% | size, $n$ |
| | 1.0000 | - | - | - | 4 |
| | 0.9000 | 1.0000 | 1.0000 | - | 5 |
| | 0.8286 | 0.8857 | 0.9429 | 1.0000 | 6 |
| | 0.7143 | 0.7857 | 0.8929 | 0.9286 | 7 |
| | 0.6429 | 0.7381 | 0.8333 | 0.8810 | 8 |
| | 0.6000 | 0.7000 | 0.7833 | 0.8333 | 9 |
| | 0.5636 | 0.6485 | 0.7455 | 0.7939 | 10 |
| | 0.5364 | 0.6182 | 0.7091 | 0.7545 | 11 |
| | 0.5035 | 0.5874 | 0.6783 | 0.7273 | 12 |
| | 0.4835 | 0.5604 | 0.6484 | 0.7033 | 13 |
| | 0.4637 | 0.5385 | 0.6264 | 0.6791 | 14 |
| | 0.4464 | 0.5214 | 0.6036 | 0.6536 | 15 |
| | 0.4294 | 0.5029 | 0.5824 | 0.6353 | 16 |
| | 0.4142 | 0.4877 | 0.5662 | 0.6152 | 17 |
| | 0.4014 | 0.4716 | 0.5501 | 0.5996 | 18 |
| | 0.3912 | 0.4596 | 0.5351 | 0.5842 | 19 |
| | 0.3805 | 0.4466 | 0.5218 | 0.5699 | 20 |
| | 0.3701 | 0.4364 | 0.5091 | 0.5558 | 21 |
| | 0.3608 | 0.4252 | 0.4975 | 0.5438 | 22 |
| | 0.3528 | 0.4160 | 0.4862 | 0.5316 | 23 |
| | 0.3443 | 0.4070 | 0.4757 | 0.5209 | 24 |
| | 0.3369 | 0.3997 | 0.4662 | 0.5108 | 25 |
| | 0.3306 | 0.3918 | 0.4571 | 0.5009 | 26 |
| | 0.3242 | 0.3848 | 0.4497 | 0.4915 | 27 |
| | 0.3180 | 0.3755 | 0.4401 | 0.4828 | 28 |
| | 0.3118 | 0.3685 | 0.4325 | 0.4749 | 29 |
| | 0.3063 | 0.3624 | 0.4251 | 0.4670 | 30 |